Human Nutrition

Nutrition

↳ Food
  ↳ Development, Growth
  ↳ Cell energy —

• Process of Nutrition

- Ingestion
- Digestion
- Assimilation
- Excretion

Nutrient

Macro Nutrient | Micro Nutrient

Macro Nutrient
↳ Large amount
↳ C → carbohydrate
↳ Protein
↳ Lipid

Micro Nutrient
↳ vitamins
↳ minerals

• Carbohydrate

↳ Carbs → carbon
↳ Hydrate → Water

(This compound contains C, H, O. elements.)

Ratio C:H:O → 1:2:1

1gm Carbohydrate = 4.2 calories.

• carbohydrate
↳ Digestion
↳ mouth
↳ Saliva
↳ Enzyme

• Carbohydrate को तोड़ने का कार्य करते है

Per day

↳ 350gm - 400gm minimum amount

carbs → mouth → saliva → ptylin Enzyme → Break down of carbs

↳ Partial digestion

Small intestine → Pancrease → Amylase Enzyme

↳ Complete digestion

Unit of carbs

↳ Saccharide

carbohydrate
→ Mono Saccharide
→ Oligo Saccharide
→ Poly Saccharide

Mono Saccharide

↳ Single Sugar unit
(single unit of Saccharide)

↳ Smallest carbohydrate

↳ $C_3 - C_7$

C_3 → Triose Sugar

C_4 → Tetrose Sugar

C_5 → Pentose

C_6 → Hexose

C_7 → Heptose

Ose → carbs

Ase → Enzyme

• Glucose ———— ($C_6H_{12}O_6$)

↳ Blood Sugar

↳ fruits, juice

↳ $C_6H_{12}O_6$

→ $-CHO$

Fructose :-

↳ Fruit

↳ $C_6H_{12}O_6$

↳ =O → (Keton)

↳ fruit sugar.

Galactose :-

↳ Lactose

↳ Breakdown

↳ -CHO

↳ milk product

↳ Beetroot, Almond.

↳ Instant Energy → monosaccharide.

Oligo Saccharide :-

↳ di-saccharide

↳ saccharide unit

2 - sugar → Di-saccharide

3 - sugar → Tri - "

4 - sugar → Tetra - "

5 - sugar → Penta - "

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• Lactose - ($C_{12}H_{22}O_{11}$)

↳ Glucose + Galactose

↳ milk Sugar.

$$\text{Lactose} \xrightarrow[\text{Bacteria}]{\text{Lactobacillus}} \text{Lactic Acid}$$

• Maltose :- ($C_{12}H_{22}O_{11}$)

↳ Glucose + Glucose

↳ Dry fruits

Poly - Saccharide

(Storage form)

• Glycogen :- (Human body)

• Starch :- (Rice, potato, corn)

• Inulin :- (plants, Dehlia)

Structural form

↳ Plant

↳ cell wall

↳ cellulose

Glycogen :- Glucose.

Starch

Insulin :- Fructose

cellulose :- β - Glucose

Chitin → NAG (N-Acetyl Glucosamine)

~~Swe~~

Sucrose - $C_{12}H_{22}O_{11}$

↳ Glucose + Fructose

↳ Jaggery

other name

Table Sugar

Honey Sugar

• Universe

↳ more

↳ cellulose

Protein

→ Organic Compound

→ Energy

→ Mulder/mulder → descoverer

→ Naming → Berzelius

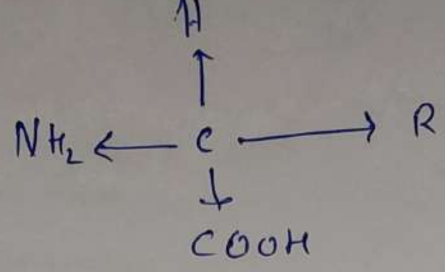
→ Structural formula → pauling

→ unit → Amino Acid

→ 20 Amino Acid

• AA -

Chiral carbon



Nature: Amphoteric

Function

- Repair work for wear and tear
- Energy
- Body building
- Wound healing
- Building block of life

• Types of Amino Acid

• Natural Amino Acid.

$$H^+ = OH^-$$

Example :-

i) Alanine

ii) Glycine

iii) Valine

2. Alcoholic AA:

↳ (OH^-) → present
↳ Serine → serine

3) Acidic AA:

- PH ↓ 7
↳ Glutamic

4) Basic AA.

- PH ↑ 7
↳ Lysine
(जूँ के अंदर पाया जाता है)

5) Aromatic AA:

↳ Hormone बनाने में

- Tyrosine
- Melanine
- Thyroxine

Amino Acid

Essential
↓ 9
Diet

Non-Essential
↓ our body

Sources

- Milk (casion) - दूध
- Soyabean - सोयाबीन
- wheat (Glutenine) - गेहू
- Egg (Albumin) - अंडा
- mushroom (Pleurotus) - मशरूम
- Pulses (Proleminus) - दाले.

(In hair, nails, hooves).

- Keratine
- Dead cells control

Eyes ↗ Opsin (color)

↓ vit. A

Eyes ↘ Rodosin (अंधेरे)

Opsin + vit A → Rodopsin

Basmati Rice

- lysine
- Pyrrole (smell) → खुशबू

Digestion : Stomach

- Pepsin Enzyme → Protein
  - Protein

Disease

P↓

1. kawashiorkar (1-2)
2. marasmus (1-2)

P↑

↳ NH_2

↳ NH_3

↳ Toxic → kidney

damage

- Hair loss
- हाथ - पैर · पतले
- पेट - फूला